

Ivonne Berenice Lemus Martínez | Embedded Software Engineer

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Embedded Software Engineer with over 5 years of experience in developing and executing complex embedded software projects. Proficient in **real-time** operating systems (**RTOS**) and **low-level** programming, with a strong knowledge of communication protocols including **CAN**, **I²C**, **SPI**, and **UART**. Demonstrated expertise in utilizing **AUTOSAR** architecture throughout the **automotive** software development life cycle. Experienced in conducting rigorous testing of secure and critical software systems, ensuring reliability and compliance with **automotive** industry standards.

PROFESSIONAL EXPERIENCE

Embedded Software Engineer - BorgWarner. 19-03-2024 – 09-06-2024

- Integrated **pull request** code **updates** into the project's main branch using **version control tools** such as **Bitbucket**, **Git**, **GitHub**, **Plastic**, and **SmartGit**.
- Ensured the proper startup of the **bootloader** and application by utilizing **CANoe** and **CANalyzer**, verifying part numbers to confirm correct software deployment.
- Collaborated with a **global cross-functional team** to develop customized solutions that resolved customer needs.

Embedded Software Engineer - CIDEA. 16 - 06 2021 - 04-12- 2023

- Integrated and configured the **watchdog driver** within the **AUTOSAR** Hardware Abstraction Layer (HAL) to detect and respond to software malfunctions during **C code** execution.
- Designed a **C**-based software component within the **AUTOSAR** architecture to monitor the **ECU** (Electronic Control Unit) circuit board temperature.
- Improved **ECU** reliability by resolving over 10 code **issues** using analysis and **debugging** tools such as **Ozone** and the **JTAG** interface to identify root causes.
- Performed software validation by transmitting **CAN** protocol messages to a **test bench** and analyzing **diagnostic** feedback to ensure compliance with the detailed software design.
- Provided **mentorship** to engineers in performing low-level testing, unit testing, and static analysis for safety-critical systems.
- Completed tasks on schedule, including software design **testing**, software requirements analysis, and **peer code reviews**, using **Jira** and **Polarion** management tools.
- Conducted **cybersecurity** testing to ensure the correct generation and secure storage of cryptographic keys within the Hardware Security Module (HSM) of the main Electronic Control Unit (ECU).
- Implemented **embedded software** in accordance with the V-Model of the **ASPICE (Automotive SPICE)** process to fulfill the **Software Development Life Cycle**.
- Performed software **unit tests** with VectorCAST to verify the detailed software design, ensuring high-quality software delivery and full condition coverage of functions.

Project Engineer - CIDESI. 12-04-2019 -14-06-2021

- Developed software in **Embedded C** language for **ARM microcontrollers** to store data in external NAND flash **memory**, improving storage speed by 80% in a glider datalogger system.
- Maximized the efficiency of TMS570LS3137 **microcontroller** system up to 30%, by developing **embedded software** implementing **RTOS (Real Time Operating System)**, using **GIT** for program version control.

- Visited customer and internal locations to provide hands-on support to **systems engineers**, identify **hardware** requirements, and contribute to the digitalization of the existing railway driving system in Mexico City.
- Developed embedded software in **C++** to create an interface for robot control, enabling abstraction and modularity through organized access to microcontroller peripherals.
- Implemented **C++ software** using Object-Oriented Programming (**OOP**) to control the speed and acceleration of Mexico City's railway driving system, enabling project flexibility and scalability.
- Performed reverse engineering on more than 20 analog amplifier circuit PCBs by **analyzing** their design and validating signals using an **oscilloscope**.
- Visited **customer** and internal locations to provide hands-on support to **systems engineers**, identify **hardware** requirements, and contribute to the digitalization of the existing railway driving system in Mexico City.
- Performing **software testing** on train tracks and troubleshooting during system start-up to verify correct implementation of the system update.
- Developed **low-level drivers** for **I²C**, **CAN**, **SPI**, & **UART** communication protocols, supporting **debugging** and validation activities, including **troubleshooting** issues.
- Implemented code to read data from digital sensors using **Python** and **embedded Linux**, optimizing GPIO (General Purpose Input/Output) usage by 50%.

Graduate Studies- Instrumented equipment Department - CIDESI. 27-09-2017 – 15-12-2020

- Implemented a **PID** in Embedded C to control the position of a robotic arm driven by a DC motor.
- Conditioned** signals from 16 Hall sensors for acquisition by an 8-bit resolution ADC by amplifying the original signals up to 10 times.
- Implemented signal processing software with a DAQ system to process the output from a DC/AC inverter, applying **filtering** techniques to smooth the waveform and extract the target signal.
- Created an electronic **architecture** for signal acquisition of 80 Hall sensors implemented in **FPGA**, which allowed to get 400 µs in acquisition time and reduce by 70% the implementation costs.

Internship - Instrumentation Equipment Department - CIDESI. 27- 04-2017 - 15-09-2017

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed a virtual instrumentation applying least squares **algorithms** to model the dynamic control of a DC motor using **Embedded C** language on a **TI microcontroller**.

EDUCATION

B.S. in Mechatronics Engineering | 09 -02-2012 - 10-02-2017

National Technological Institute of Mexico | San Luis Potosí

- Developed **firmware** for Texas Instruments, Microchip, and ST **microcontrollers** with **Assembly** and **C** languages.

M.Sc. & T. Mechatronics | 07-09-2017 - 09-12-2019

CIDESI | Querétaro

- Developed and executed **embedded** software to test a **hardware architecture** for analog data acquisition.

SKILLS

Programing Languages: C, C++ & Assembler

DevOps: Git & Plastic

CPU architectures: FPGA (Artix-7), FPGA (Spartan 3E), TMS570LS3137, MSP430F2618, PIC4550, STL microcontroller.

Communication protocols: I2C, SPI, UART & CAN

Operating systems: RTOS, SafeRTOS & FreeRTOS

Languages: English & Spanish

COURSES

C++: UDEMY. 2024-09-10 - 2024-10-10

Vector tools: CIDEDEC. 2021-06-16 - 2021-09-27